

Week 4 — Sampling & the CLT

Business Forecasting

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Where we are

C1 Framing → C2 Describing data → **C3 Uncertainty** → C4–C5 Regression → C6 Forecasting

We enter **C3 — uncertainty**: our data is only a *sample*, so any number we compute is a *guess*.

- This week (W4): the **random sample**, the **estimator** and its properties (**bias, variance, MSE**), the **sampling distribution**, and the **Central Limit Theorem**.
- Next week (W5) turns this into **confidence intervals**.

This week: review, done fast in R

- You have met these ideas before — we move quickly and lean on **simulation in R** to see them.
- The CLT especially is best *simulated*: draw many samples, watch the distribution of the mean take shape.
- Practice as we go with `W04_learnr.Rmd`.

Motivating example

You run a bunch of **Airbnbs** in Mexico City. Should you invest more in cleaning?

- Concretely: can you charge a higher price if your cleanliness score exceeds **4.5**?
- Get a sample of listings and compare the price of
 - those with cleanliness score below 4.5 (**dirty**)
 - and above 4.5 (**clean**)

Motivating example: the sample

We have a sample of 200 listings (`price` , cleanliness `clean`):

id	review_scores_cleanliness	price	clean
40032982	3.00	1023	Dirty
21962322	4.50	4500	Dirty
41841538	4.50	380	Dirty
624813934659858771	3.40	1350	Dirty
47030021	4.29	684	Dirty
46736994	4.29	451	Dirty

Statistical language: population & parameter

- **Population:** the entire group we want to learn about — impossible to assess directly.
 - All Airbnb listings in Mexico City; ideally the whole distribution of prices.
- **Parameter:** a number describing a characteristic of the population.
 - The mean price of clean μ_c and dirty μ_d apartments.

► `learnr W04_learnr.Rmd` → Exercise 2: The population parameter.

Statistical language: sample & goal

- **Sample:** the part of the population we have data for.
 - Our 200 listings.
- **Goal:** what we want to learn about the population.
 - Is $\mu_c > \mu_d$? If so, by how much?
 - We don't know μ_c, μ_d — we **guess** them with an **estimator** and a random IID sample.

What is a random sample?

- **At random:** each member of the population has an equal chance of being selected. Selecting is called *drawing* a sample.
- **Random variable P_i :** describes observation i *before* drawing — it could be any price from the distribution.
- **Random sample:** a collection of random variables $\{P_1, P_2, \dots, P_n\}$.
- **Observed value p_i :** once realized, it is a constant from our sample — no longer random.

Before drawing the sample

The P_i are still random — no values yet:

Random Variables P_i						
	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs						
Realized Values p_i						

Random sample – Draw 1

Drawing realizes each P_i into a concrete value p_i :

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	8451	9015	8161	9085	8268	1622
Realized Values p_i	120	150	800	200	1400	110

Random sample — Draw 2

A different draw from the **same** population gives **different** realized values:

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	1933	3947	3145	3773	6721	3373
Realized Values p_i	260	420	500	2120	800	1450

The point estimate changes **every** time we draw.

IID samples

- IID = Independent and Identically Distributed.
- **Independent:** selecting one unit P_i doesn't affect selecting another P_j .
- **Identically distributed:** all units P_i come from the same distribution.

Estimators: intuition

- Our **method of guessing** the parameter from the data.
- A function of the random variables in the sample: $\hat{\theta} = f(P_1, \dots, P_n)$.
- Because it is random, we can analyze its statistical properties.
- It **cannot** contain unknown quantities (like σ or μ).

Estimators: ones we've seen

- Sample mean: $\hat{\mu}_c = \bar{P} = \frac{\sum_n P_i}{n}$
- Sample standard deviation: $s_c = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (P_i - \bar{P})^2}$
- Each is a rule that turns a sample into a single guess.

Point estimate

- A single number computed from the realized data $\{p_1, \dots, p_n\}$:

$$\bar{p} = f(p_1, \dots, p_n) = \frac{\sum_n p_i}{n}$$

- No longer random — it is one number from one sample.

One number isn't enough

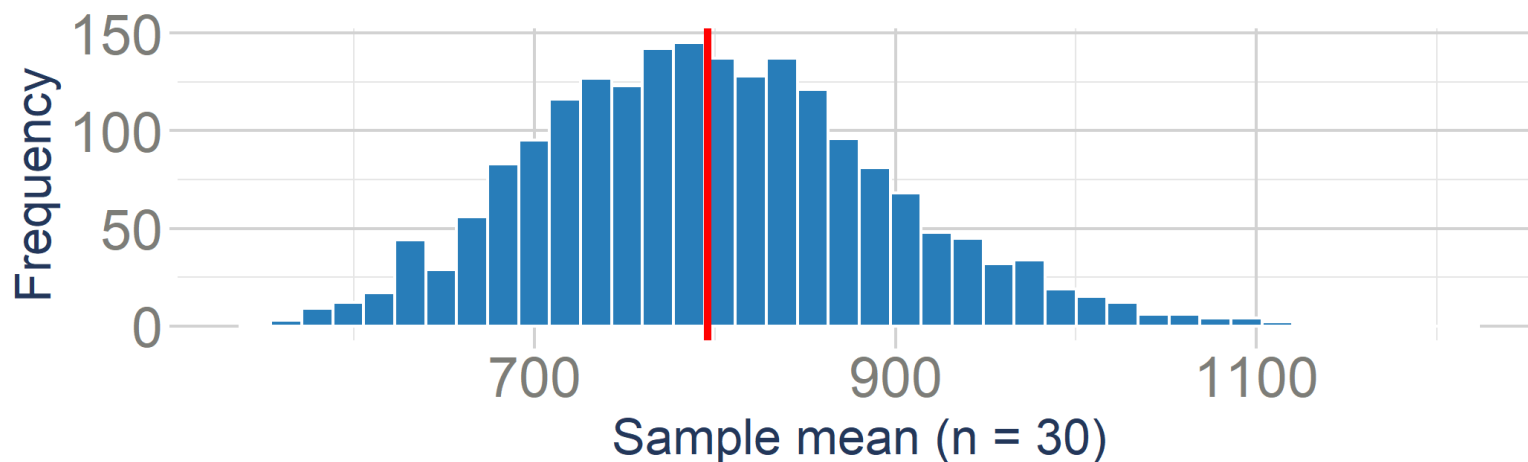
- The mean price of clean listings in our sample is $\bar{p}_c = 1245.4$ MXN — our point estimate.
- We **can't** say how close this one number is to the true mean without knowing the population.
- But we **can** judge our *method* (the estimator) by its **sampling distribution**.

► [learnr W04_learnr.Rmd](#) → Exercise 1: One sample, one statistic.

Simulate the sampling distribution

Draw 2,000 samples of $n = 30$ from a skewed price population and plot the means.

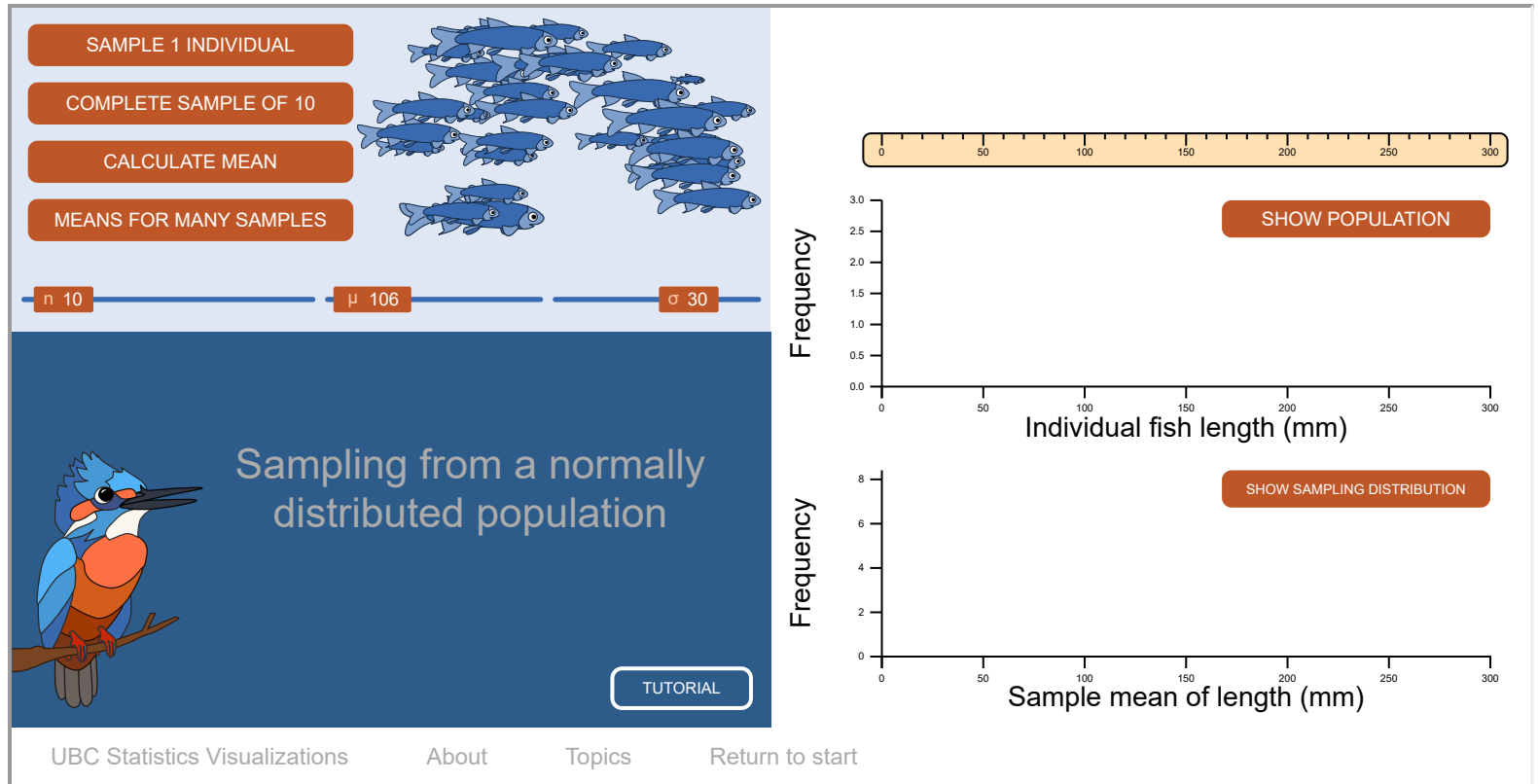
```
set.seed(1)
population <- rlnorm(100000, meanlog = 6.5, sdlog = 0.6) # skewed price
means <- replicate(2000, mean(sample(population, 30)))
ggplot(data.frame(means), aes(means)) +
  geom_histogram(bins = 40, fill = "#2980b9", color = "white") +
  geom_vline(xintercept = mean(population), color = "red", linewidth = 2) +
  labs(x = "Sample mean (n = 30)", y = "Frequency") + theme_xaringan()
```



The means cluster around the **true mean** (red) — the estimator has a *distribution*.

Sampling distribution: explore it

Each estimator has its own sampling distribution — try a few:



From sampling distribution to the CLT

- We know the **mean** and **variance** of the sample mean. Can we say anything about its **whole distribution**?
- For the sample mean — **yes!**
- That's the **Central Limit Theorem**, the most exciting theorem in statistics.

CLT: the setup

$X_1, \dots, X_n$ are i.i.d. with mean μ and st. dev. σ .

- Sum $S_n = \sum_n X_i$: $E[S_n] = n\mu$, $sd(S_n) = \sqrt{n} \sigma$.
- Mean $\bar{X}_n = \frac{S_n}{n}$: $E[\bar{X}_n] = \mu$, $sd(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$.
- Standardized $Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$: $E[Z_n] = 0$, $sd(Z_n) = 1$.

CLT: the theorem

For large n :

$$S_n \dot{\sim} \mathcal{N}(n\mu, \sqrt{n}\sigma), \quad \bar{X}_n \dot{\sim} \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right), \quad Z_n \dot{\sim} \mathcal{N}(0, 1)$$

- In large samples $\bar{X}_n$ is **approximately normal**, centered at μ , with st. dev. $\frac{\sigma}{\sqrt{n}}$.
- As n grows, the approximation improves and the spread **shrinks with $\sqrt{n}$** .

► [learnr W04_learnr.Rmd](#) → Exercise 5: The standard error.

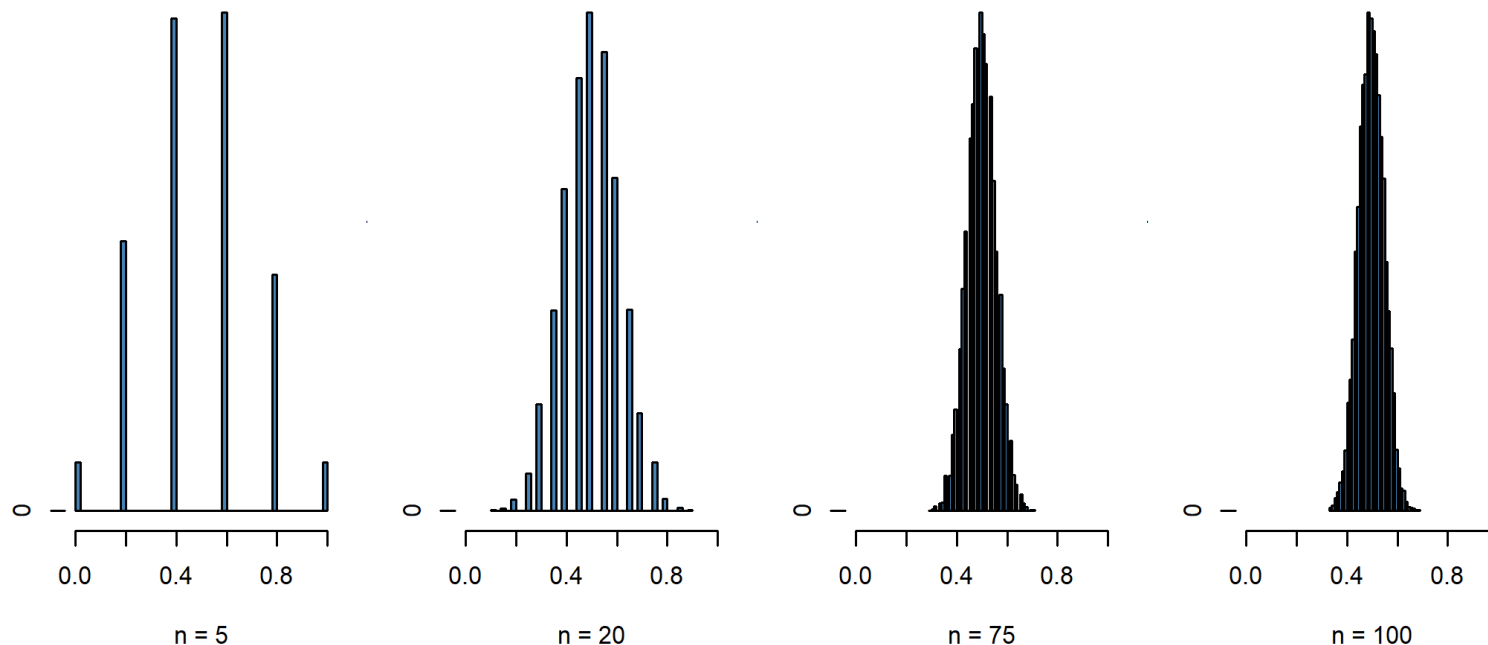
CLT: what drives the shape

- The **original** distribution of X_i doesn't matter (but outliers slow convergence).
- Larger $n \rightarrow$ tighter distribution around the mean.
- Smaller $\sigma \rightarrow$ tighter distribution around the mean.

► [learnr W04_learnr.Rmd](#) → Exercise 8: As n grows....

CLT: simulate it (even for discrete data)

$X_i \sim \text{Bernoulli}(0.5)$. Distribution of $\bar{X}_n$ for growing n — it becomes normal and **tighter**:



- St. dev. of the mean: $\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} = \frac{\sqrt{p(1-p)}}{\sqrt{n}} = \frac{0.5}{\sqrt{n}}$.

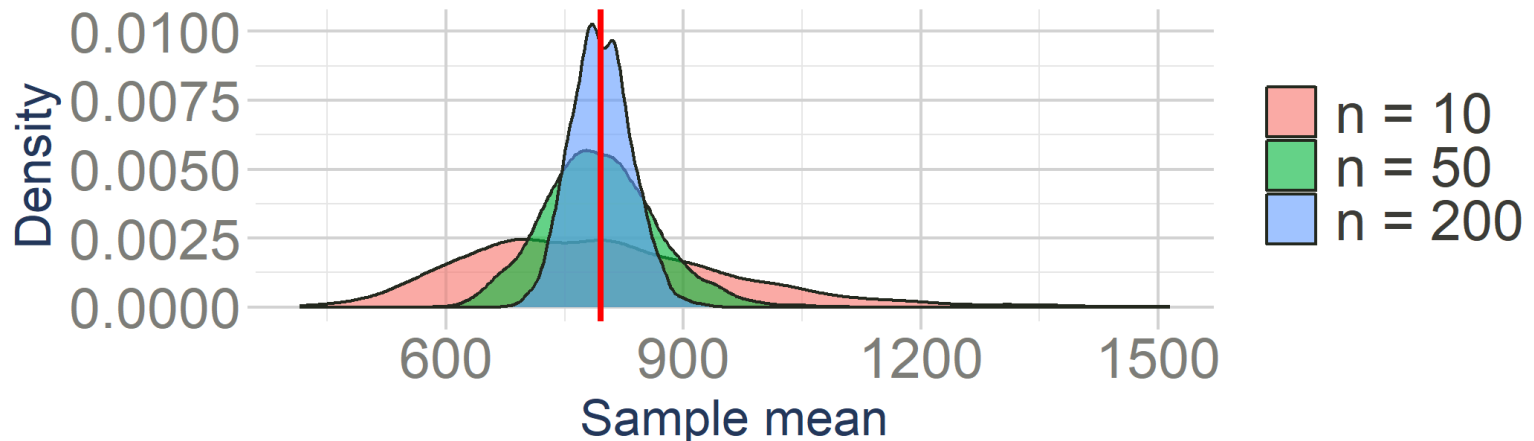
► [learnr W04_learnr.Rmd](#) → Exercise 6: The CLT — skewed population, bell-shaped means.

The standard error shrinks with $\sqrt{n}$

As n grows the sampling distribution stays centered at μ but gets **tighter**: $sd(\bar{X}_n) = \sigma/\sqrt{n}$.

► [learnr W04_learnr.Rmd](#) → Exercise 7: Bigger n, tighter distribution.

```
set.seed(7)
pop <- rlnorm(100000, meanlog = 6.5, sdlog = 0.6)
df <- do.call(rbind, lapply(c(10, 50, 200), function(n)
  data.frame(mean = replicate(1500, mean(sample(pop, n))), n = paste0("n = ", n))
df$n <- factor(df$n, levels = c("n = 10", "n = 50", "n = 200"))
ggplot(df, aes(mean, fill = n)) +
  geom_density(alpha = 0.6) +
  geom_vline(xintercept = mean(pop), color = "red", linewidth = 1) +
  labs(x = "Sample mean", y = "Density", fill = NULL) + theme_xaringan()
```



Normal distribution: combining variables

A customer who opens the DiDi app may call a car. X = calls in Cancún (Bernoulli 0.4), Y = in Puerto Vallarta (Bernoulli 0.6). Sample: 100 from Cancún, 80 from PV. **P(more than 100 people call a car)?**

Reminders (for normal X):

$$X + c \sim \mathcal{N}(\mu + c, \sigma), \quad cX \sim \mathcal{N}(c\mu, |c|\sigma)$$

$$X + Y \sim \mathcal{N}\left(\mu_1 + \mu_2, \sqrt{\sigma_1^2 + \sigma_2^2}\right)$$

► `learnr W04_learnr.Rmd` → Exercise 6: The CLT — skewed population, bell-shaped means.

What if I don't know σ ?

Store sales are normal with mean 200 and **unknown** variance. Take $n = 80$ stores: $P(\bar{X} > 220)$?

- By the CLT, $\bar{X} \sim \mathcal{N}\left(200, \frac{\sigma}{\sqrt{80}}\right)$ — but σ is unknown.
- Use the **sample** standard deviation to estimate σ .
- That adds a little uncertainty, but in a large sample you estimate σ well.
- The distribution still converges to normal — you just need a few more observations.

EXAM QUESTION — Fall 2025 Midterm 1

The sampling/estimation logic here is the **foundation** for the Midterm's inference questions.

Last year's Midterm 1 CI questions (Q7 / Q8) build directly on the **sampling distribution of the mean** and $SE = \sigma / \sqrt{n}$ — but they ask for **confidence intervals**, which we build in **Week 5**. This week, make sure the *estimator + sampling distribution* logic is second nature. **We solve the full CI questions together next week.**

What you must be able to do this week

- Tell a **random variable** P_i from a **realized value** p_i , and state what makes a sample IID.
- Write an **estimator** as a function of the sample and compute a **point estimate**.
- Check an estimator for **bias**, compare two by **variance/efficiency**, and combine both via $\text{MSE} = \text{Var} + \text{Bias}^2$.
- Describe the **sampling distribution** of the mean and **simulate** it in R.
- State the **CLT**: $\bar{X}_n \approx \mathcal{N}(\mu, \sigma/\sqrt{n})$ — approximately normal, centered at μ , spread **shrinking with** $\sqrt{n}$.

Next week (W5): turn the sampling distribution into **confidence intervals**.

Appendix — optional theory refresher

Example: the estimator (before drawing)

We want the average price in Mexico City but lack the whole population. Take a sample of 8 listings and average.

Random Variables P_i						
	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs						
Realized Values p_i						

Estimator: $\hat{\mu} = \frac{P_1 + \dots + P_8}{8}$

Example: the estimator (Sample 1)

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	8451	9015	8161	9085	8268	1622
Realized Values p_i	120	150	800	200	1400	110

Point estimate (all 8): $\frac{p_1 + \dots + p_8}{8} = 685$

Example: the estimator (Sample 2)

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	1933	3947	3145	3773	6721	3373
Realized Values p_i	260	420	500	2120	800	1450

Point estimate (all 8): $\frac{p_1 + \dots + p_8}{8} = 809.9$ — a different sample, a different estimate.

Expectation of an estimator: bias

- A good estimator is **unbiased**: $E[\hat{\theta}] = \theta$ for any value of θ .
- Its sampling distribution is **centered** at the parameter — on average it hits the truth.

$$Bias(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- Take many samples, compute the mean in each: the average of those means should be the true mean.
- Let's check a few estimators.

► [learnr W04_learnr.Rmd](#) → Exercise 9: Unbiasedness.

Example 1: $\hat{\theta}_1 = X_i$ (a single observation)

- X_i has unknown mean $E(X_i) = \mu$; we want to estimate μ .
- Estimator: $\hat{\theta}_1 = X_i$.
- Expected value: $E(\hat{\theta}_1) = E(X_i) = \mu$.
- Bias: $E(\hat{\theta}_1) - \mu = 0 \rightarrow$ unbiased.
- Is it a *good* estimator? (We'll come back to this.)

► learnr W04_learnr.Rmd → Exercise 4: Centered on the parameter.

Example 2: $\hat{\theta}_2 = (3X_1 + X_2)/5$

- Same setup: $E(X_i) = \mu$.
- Expected value: $E(\hat{\theta}_2) = \frac{3}{5}\mu + \frac{1}{5}\mu = \frac{4}{5}\mu$.
- Bias: $\frac{4}{5}\mu - \mu = -\frac{1}{5}\mu \rightarrow$ **biased** (weights don't sum to 1).

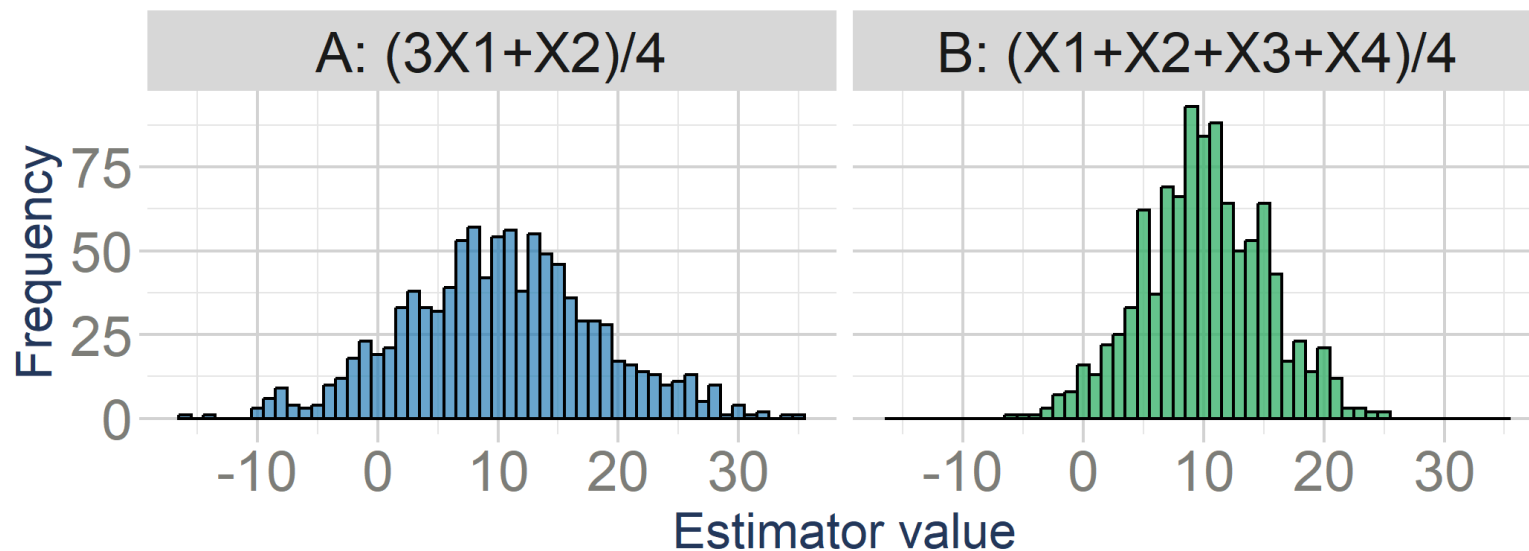
Example 3: $\hat{\theta}_3 = \bar{X} = \frac{\sum X_i}{n}$ (sample mean)

- Same setup: $E(X_i) = \mu$.
- Expected value: $E(\bar{X}) = \frac{\sum_n E(X_i)}{n} = \frac{n\mu}{n} = \mu$.
- Bias: 0 $\rightarrow$ **unbiased**. The sample mean is our workhorse estimator.

Variance of an estimator: efficiency

Among **unbiased** estimators, prefer the one with **smaller variance** (more *efficient*). Sample IID from $X \sim \mathcal{N}(10, 10)$:

- A: $\hat{\mu}_A = (3X_1 + X_2)/4$ B: $\hat{\mu}_B = (X_1 + X_2 + X_3 + X_4)/4$ (both unbiased)



B is tighter — more efficient.

Variance: definition & relative efficiency

$$Var(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$$

- Lower variance = more efficient (a property of *unbiased* estimators). Here:

$$Var(\hat{\mu}_A) = \frac{10\sigma^2}{16} > \frac{4\sigma^2}{16} = Var(\hat{\mu}_B)$$

- Relative efficiency: $Eff(\hat{\mu}_B, \hat{\mu}_A) = \frac{Var(\hat{\mu}_A)}{Var(\hat{\mu}_B)} = \frac{10}{4} = 2.5$ — B is 2.5× as efficient.

Variance: worked examples

- $\hat{\theta}_1 = X_i: Var(\hat{\theta}_1) = \sigma^2$
- $\hat{\theta}_3 = \bar{X} = \frac{\sum X_i}{n}: Var(\hat{\theta}_3) = \frac{\sigma^2}{n}$ — shrinks with n .
- $\hat{\mu}_A = \frac{3X_1 + X_2}{4}: Var(\hat{\mu}_A) = \frac{9\sigma^2 + \sigma^2}{16} = \frac{10\sigma^2}{16}$

Side note: what if variables aren't independent?

So far we assumed independence. Suppose X_1, X_2 are **not** independent (e.g. daily sales of two products in one store).

- $E(X_1 + X_2) = E(X_1) + E(X_2)$ (always true).
- $Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2 Cov(X_1, X_2)$
- $Var(X_1 - X_2) = Var(X_1) + Var(X_2) - 2 Cov(X_1, X_2)$

Mean Squared Error (MSE)

A single summary of how good an estimator is:

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

- The **lower** the MSE, the better. It combines **bias** and **variance**.

MSE = Variance + Bias²

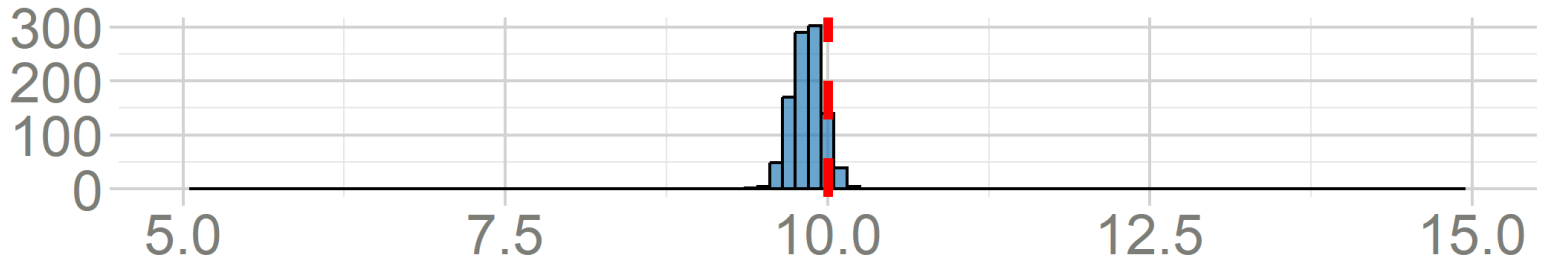
$$\begin{aligned}MSE(\hat{\theta}) &= E[(\hat{\theta} - \theta)^2] = E[(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2] \\&= E[(\hat{\theta} - E(\hat{\theta}))^2] + 2\underbrace{E[\hat{\theta} - E(\hat{\theta})]}_{=0}(E(\hat{\theta}) - \theta) + (E(\hat{\theta}) - \theta)^2 \\&= \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2\end{aligned}$$

- If unbiased, $MSE(\hat{\theta}) = \text{Var}(\hat{\theta})$.

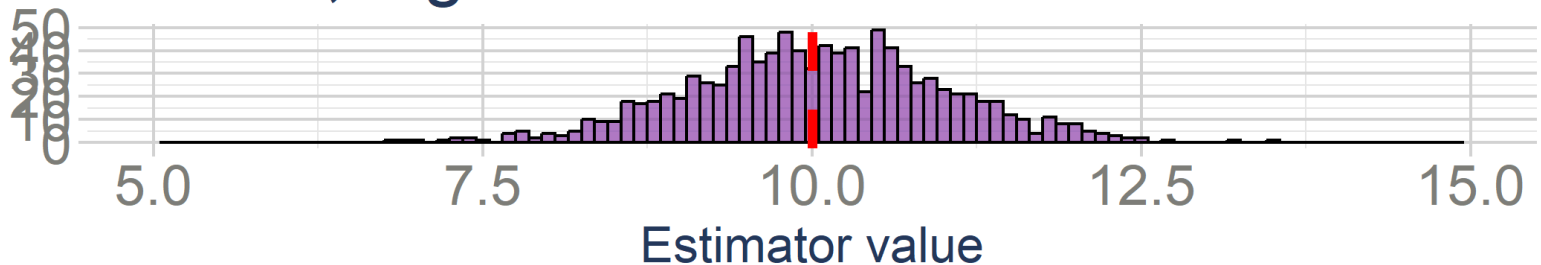
Trading bias for variance

Estimating customer income. Red line = true value. Which estimator would you prefer?

Biased, low variance



Unbiased, high variance

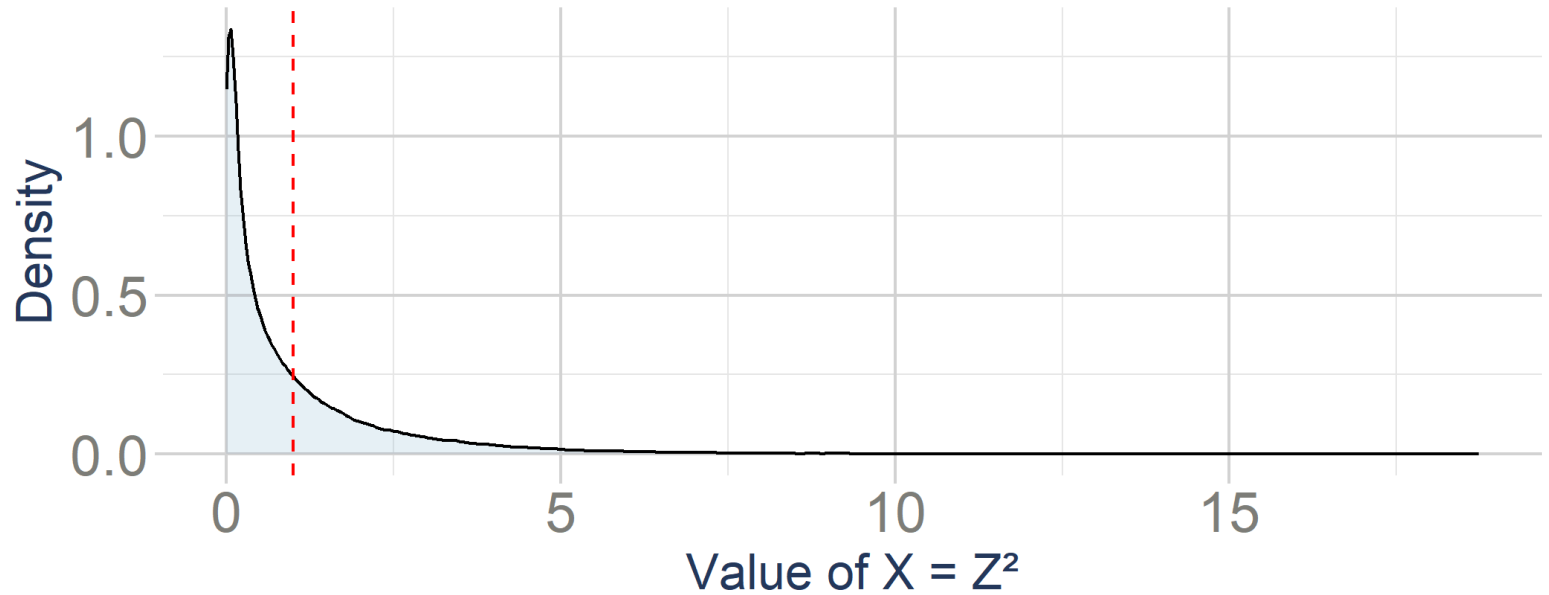


Sampling distribution of the standard deviation

- Sample **means** are normal in large samples. What about the sample **standard deviation**?
- If we take many samples and compute each one's SD, what does *that* distribution look like?
- If X_i is normal, the sample variance is related to the **chi-squared** distribution.
- This is what we'll use to build confidence intervals for the variance.

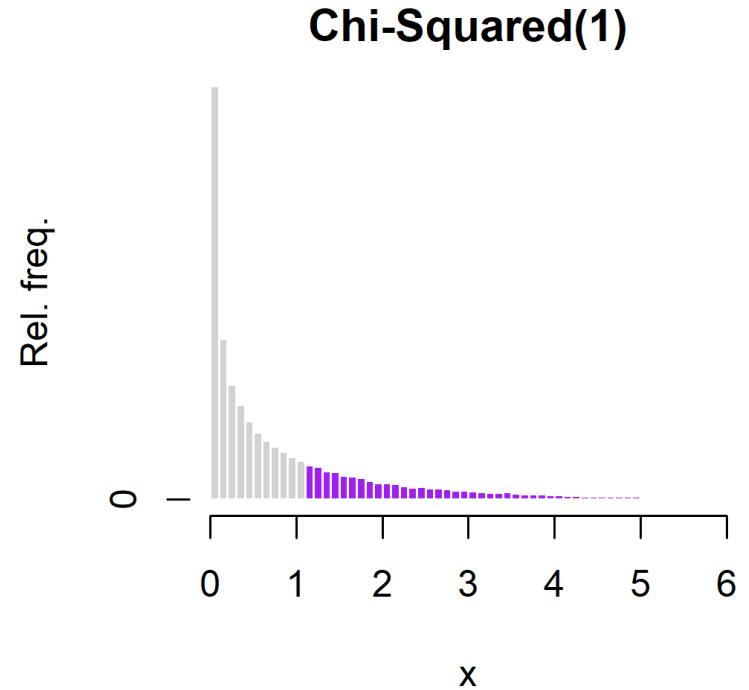
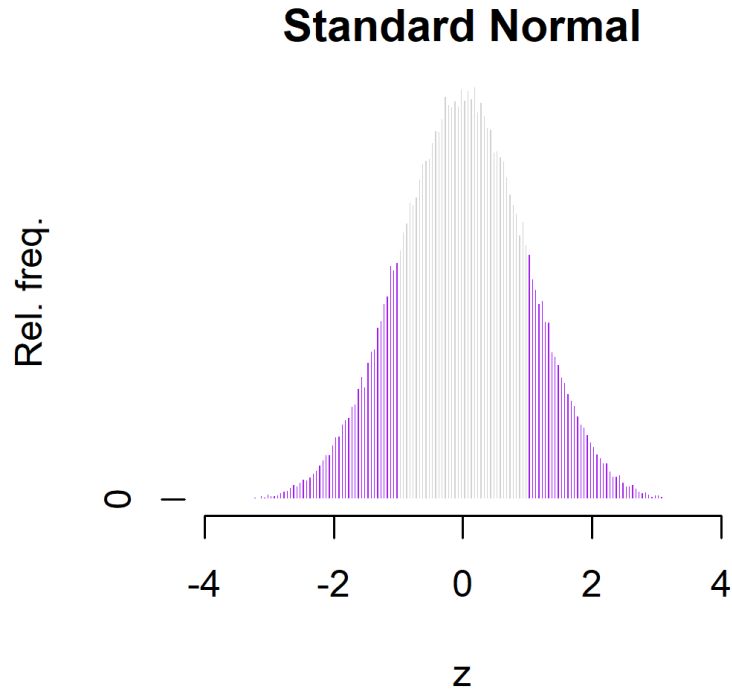
From Normal to Chi-Square

- Start with the standard normal $N(0, 1)$. The transformation $X = Z^2$ gives $\chi^2(1)$.
- $E[\chi^2(1)] = Var(Z) = 1; Var(\chi^2(1)) = 2$.



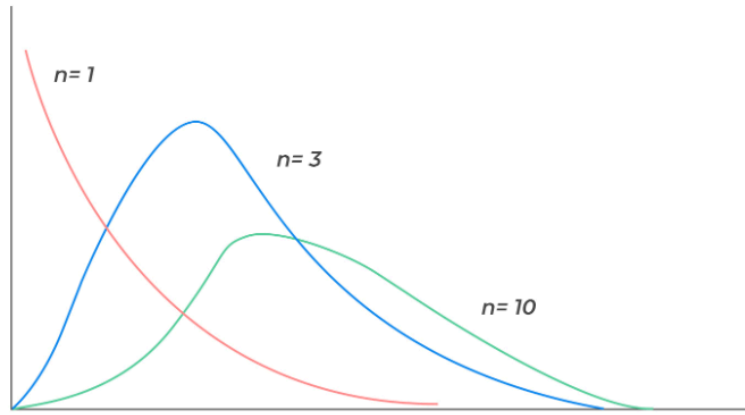
Visualizing the connection

Shaded area = $P(X = Z^2 > 1)$ where $X \sim \chi^2(1)$, $Z \sim N(0, 1)$ — same area in both panels:



Chi-Square & the sum of normals

- Sum of n i.i.d. squared standard normals is $\chi^2(n)$: $\sum_n Z^2 \sim \chi^2(n)$.
- $E[\chi^2(n)] = n$; $Var(\chi^2(n)) = 2n$.



- Why does the shape approach normal for large n ? Because it's a **sum of random variables** — the CLT again.